



US 20250283600A1

(19) **United States**

(12) **Patent Application Publication**
ZURBACH et al.

(10) **Pub. No.: US 2025/0283600 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **METHOD FOR INJECTING A
HYDROGEN-AIR MIXTURE FOR A
TURBINE ENGINE BURNER**

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(21) Appl. No.: **18/861,683**

(22) PCT Filed: **May 2, 2023**

(86) PCT No.: **PCT/FR2023/000071**

§ 371 (c)(1),

(2) Date: **Oct. 30, 2024**

(30) **Foreign Application Priority Data**

May 2, 2022 (FR) 2204146

Publication Classification

(51) **Int. Cl.**

F23R 3/28 (2006.01)

F23R 3/02 (2006.01)

F23R 3/14 (2006.01)

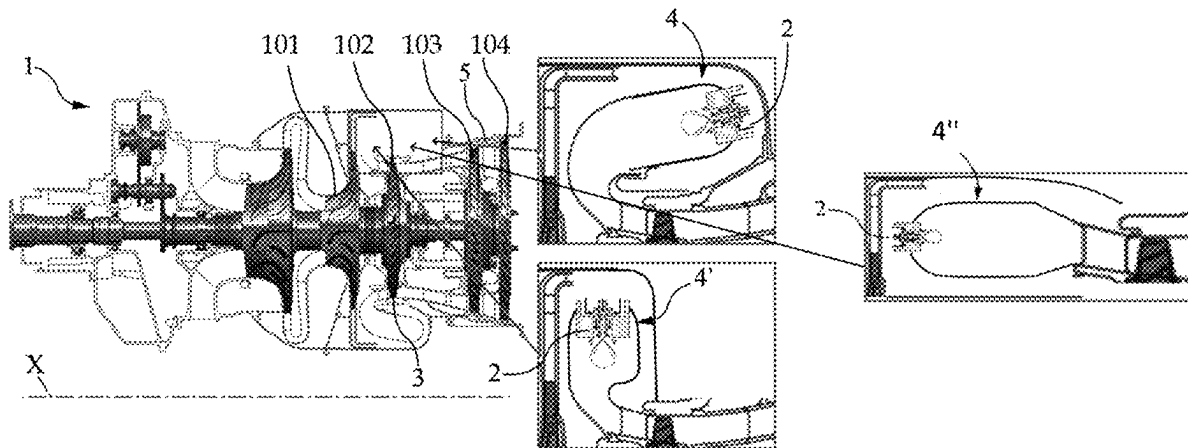
(52) **U.S. Cl.**

CPC **F23R 3/28** (2013.01); **F23R 3/02**
(2013.01); **F23R 3/14** (2013.01)

(57)

ABSTRACT

An injection method, for an injection device in a combustion chamber of an aircraft turbine engine, said injection device comprising an internal channel surrounded by an external annular channel, said channels leading into said combustion chamber of the gas turbine, the method including injecting a gaseous hydrogen/air mixture having a greater hydrogen richness level than the stoichiometric amount into said internal channel and injecting air into the external annular channel so as to produce, at the outlet of the internal channel, a first flame front resulting from rich combustion surrounded by a second flame front resulting from lean combustion.



[Fig. 1]

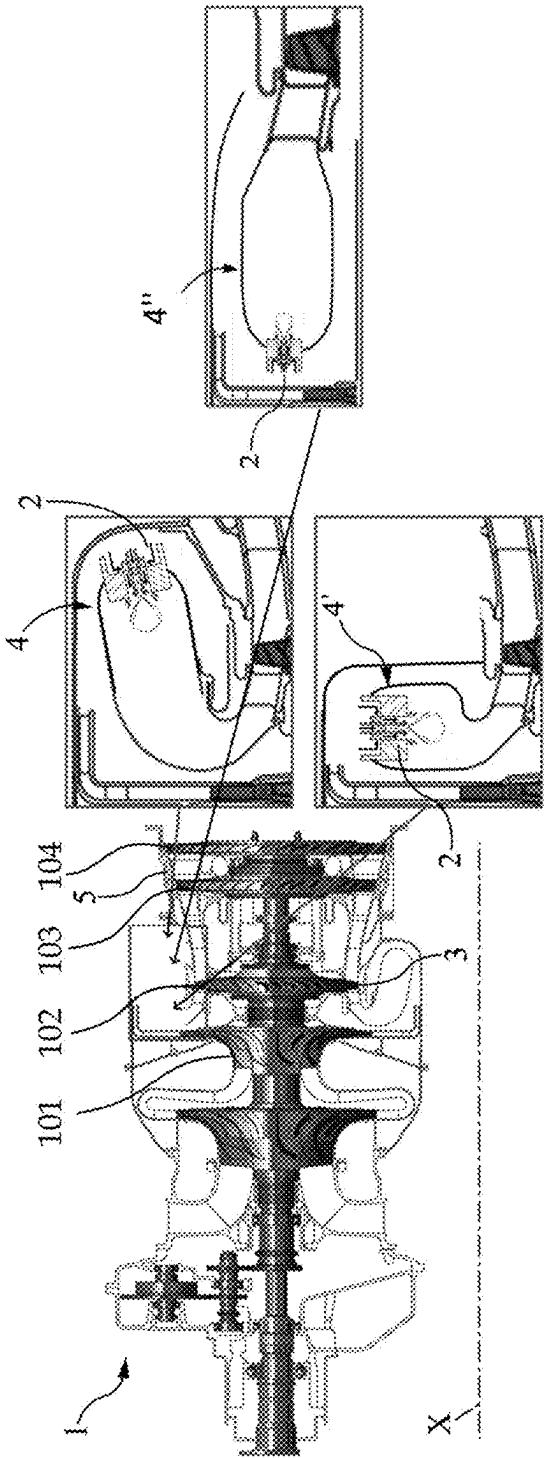
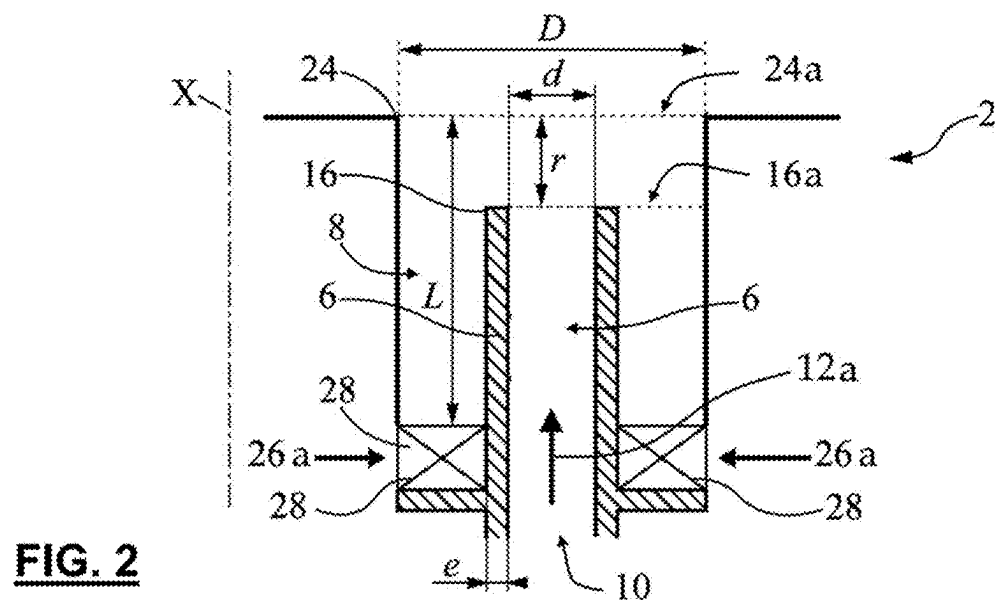
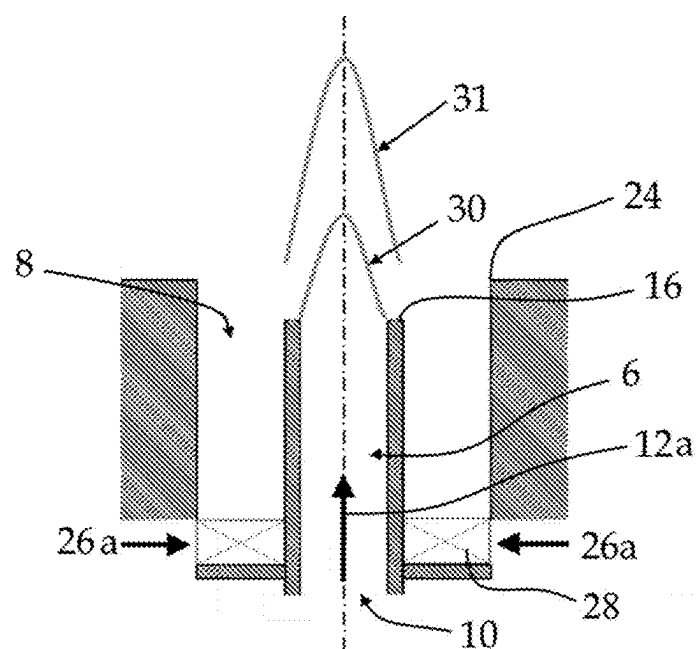


FIG. 1

[Fig. 2]



[Fig. 3]



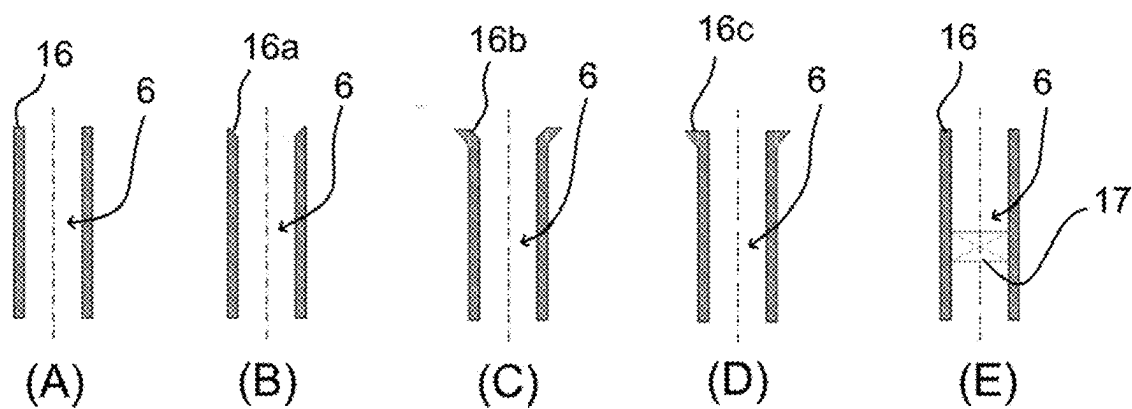


FIG. 4

[Fig. 4]

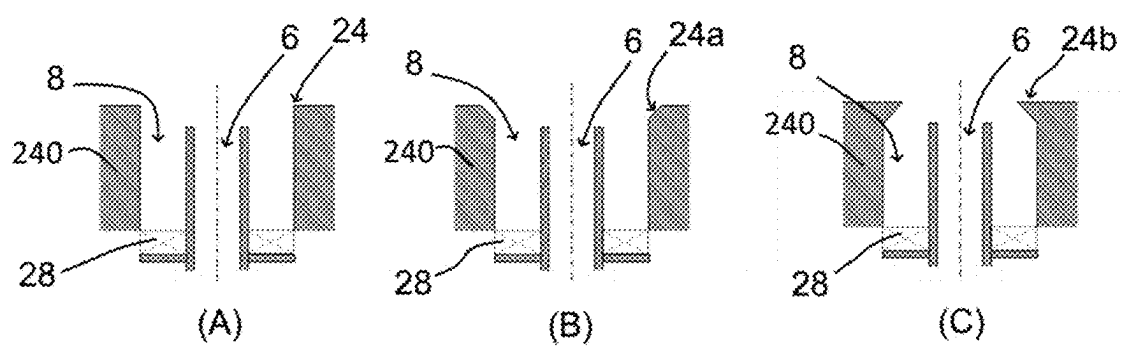


FIG. 5

[Fig. 5]

METHOD FOR INJECTING A HYDROGEN-AIR MIXTURE FOR A TURBINE ENGINE BURNER

TECHNICAL FIELD

[0001] This disclosure relates to the field of methods for supplying fuel to injection devices for gas turbines such as aircraft turbine engines powered by gaseous hydrogen and air. This includes civil and military aeronautical applications: helicopters, VTOLs, drones, APUs, turbogenerators, fixed-wing aircraft for leisure, for business, or commercial, turbojet engines, or turboprop engines.

[0002] PRIOR ART

[0003] The propulsion field, and in particular the aeronautics field, is facing major environmental challenges. There is increasingly strong interest in using combustion based on gaseous hydrogen rather than kerosene because burning gaseous hydrogen would avoid the emission of carbon pollutants such as carbon dioxide, carbon monoxide, unburned hydrocarbons, or fine particles and smoke.

[0004] A principle based on burners where air and gaseous hydrogen are micromixed is known. Burners implemented according to this principle do not guarantee the absence of flashback in the gaseous hydrogen injection device and have a complex geometry. Such burners have a high production cost, a high pressure loss, and are specific to a given combustion chamber architecture.

[0005] As for the injection and combustion, there exist two main technological configurations for hydrogen/air injection systems applied to gas turbines, namely lean injection systems and rich injection systems.

[0006] More generally, it is important to keep in mind that fuel supply methods involving lean combustion tend to generate significant thermal-acoustic instabilities that can damage these systems, while stable combustion is necessary to avoid altering engine performance. Fuel supply methods involving rich combustion, on the other hand, tend to emit more pollutants than lean combustion methods if they are not properly designed.

TECHNICAL PROBLEM

[0007] The use of hydrogen involves several issues to be taken into consideration at the combustion chamber:

[0008] Under equivalent thermodynamic conditions for pressure, temperature, and richness, the adiabatic temperature of the flame of hydrogen/air combustion is higher than the flame coming from kerosene-air combustion.

[0009] Similarly, the flame speeds from hydrogen/air combustion are higher than for kerosene-air flames. A high flame speed may lead to flashback problems in injection systems, particularly at the boundary layers, and may cause serious damage to these systems or may even cause safety issues.

[0010] The flammability range of hydrogen is, however, wider than that of kerosene and allows a hydrogen/air mixture to be ignited at lower or higher richness levels than kerosene, which ultimately can allow lower flame temperatures to be achieved than with the use of kerosene.

[0011] Finally, combustion of hydrogen with air tends to emit a lot more noise than conventional kerosene combustion and may therefore generate significant noise pollution at airports.

SUMMARY OF THE INVENTION

[0012] This document proposes a method for injecting rich premix, dedicated to the combustion of gaseous hydrogen and air, which allows addressing the technical problems presented above.

[0013] More specifically, this disclosure proposes an injection method, for an injection device

[0014] in a combustion chamber of an aircraft turbine engine, said injection device comprising an internal channel surrounded by an external annular channel, said channels leading into said combustion chamber of said gas turbine, the method comprising an injection of a gaseous hydrogen/air mixture having a greater hydrogen richness level than the stoichiometric amount into said internal channel and an injection of air into said external annular channel so as to produce, at the outlet of said internal channel, a first flame front resulting from rich combustion surrounded by a second flame front resulting from lean combustion, after ignition of the mixture. The method, in which the injection takes place continuously after ignition in order for

[0015] the turbine to operate, allows reducing the temperature of the flame fronts, which reduces the NO_x content of the gases burned and reduces wear of the injector.

[0016] The features set forth in the following paragraphs correspond to embodiments that may be implemented independently of each other or in combination with each other where appropriate:

[0017] The gaseous hydrogen/air mixture may have a hydrogen richness level that is greater than 2.

[0018] Advantageously, said gaseous hydrogen/air mixture may have a hydrogen richness level that is greater than or equal to 4.

[0019] The air flow rate in the external annular channel may be chosen such that the overall richness level at the outlet of the internal channel/external annular channel assembly is set between 0.15 and 0.5 depending on the operating points of the turbine engine.

[0020] The injection of the gaseous hydrogen/air mixture and the injection device may be configured to create, at the outlet of the internal channel, said first flame front resulting from the rich combustion of said mixture, and to attach it to a peripheral lip of the internal channel.

[0021] The hydrogen richness of the mixture may be chosen so that said rich combustion is carried out with a flame front temperature of less than 1800 K, which protects the combustion chamber.

[0022] The hydrogen richness of the mixture may be chosen so that the first flame front is laminar and has a Lewis number greater than 1, limiting diffusive-thermal instabilities and thus avoiding flashback phenomena.

[0023] The mixture burned in the first flame front generates residual gases which are advantageously burned in the second flame front which is stabilized by the supply of air from the external annular channel.

[0024] The richness of the second flame front is such that the second flame front may be maintained at a temperature below 1800 K.

[0025] The air injected by the annular channel may be rotated by an annular swirler so as to make the second flame front turbulent and so that this second flame front is not attached to the lip of the internal channel.

[0026] Advantageously, positioning the downstream end of the internal channel upstream of the downstream end of the external annular channel allows optimizing the mixing

between the gases from the first combustion and the air injected by the external channel.

BRIEF DESCRIPTION OF DRAWINGS

[0027] Other features, details and advantages of the invention will become apparent upon reading the detailed description below of some non-limiting exemplary embodiments, and upon analysis of the appended drawings, in which:

[0028] FIG. 1 shows a turbine engine comprising an injection device arranged in an annular bottom of an annular combustion chamber according to three configurations;

[0029] FIG. 2 shows a first schematic example in a cross-sectional side view of an injection device to which the method of this disclosure applies;

[0030] FIG. 3 shows a schematic view of the device of FIG. 2 in a combustion situation;

[0031] FIG. 4 shows a plurality of possible configurations (FIGS. A, B, C, D, E) of the internal channel of a device to which the method of this disclosure applies;

[0032] FIG. 5 shows a plurality of exemplary configurations of annular channel outlets (FIGS. A, B, C) for a device to which the method of this disclosure applies.

Description of embodiments

[0033] The drawings and the description below contain elements that may not only serve to provide a better understanding of the invention, but where appropriate may also contribute to its definition.

[0034] Reference is now made to FIG. 1 which shows three example configurations for the installation of an injection device 2 on a turbine engine 1, according to the orientation of the annular bottom of an annular combustion chamber 4, 4', 4'' of the turbine engine: either combustion chamber 4'' is oriented substantially along a longitudinal axis X, or combustion chamber 4 is oriented at an acute angle relative to this longitudinal axis, or combustion chamber 4' is transverse to this longitudinal axis X. In all cases, injection device 2 is installed between a 5 compressor 101 and a high-pressure turbine 102, 103, 104, on an annular bottom of annular combustion chamber 4, 4', 4'' or on an external shell.

[0035] The injection device may be, as illustrated in FIG. 2, an injection device that comprises an internal channel 6 and an external annular channel 8. External channel 8 is centered on internal channel 6, and in the case of tubular channels, internal channel 6 and external annular channel 8 are coaxial. These channels lead into combustion chamber 4, 4', 4'' of the device of FIG. 1. The internal and external channels have a circular cross-section. An ignition device (not shown) allows the gases exiting the channels to be ignited in order to initiate combustion.

[0036] Injection device 2 is used in the present disclosure in a configuration where a rich mixture of gaseous hydrogen/air is injected into internal channel 6 while air is injected into 15 external channel 8. As a result, the combustion comprises a first hydrogen-rich combustion at the outlet of internal or central channel 6 and a second "lean" combustion which is carried out around a flame created by the first combustion.

[0037] For the injection into internal channel 6 and the combustion at the outlet of this channel, the injection and combustion are said to be rich when there is excess gaseous hydrogen 20 compared to a combustion taking place at the stoichiometric ratio between the gaseous hydrogen and

gaseous oxygen in air, and the injection and combustion are said to be lean when there is excess gaseous oxygen compared to this stoichiometric combustion. Stoichiometric combustion is defined as combustion where there is the correct number of hydrogen and oxygen atoms required in order to consume all the fuel, and only water and nitrogen remain in the combustion 25 products.

[0038] According to FIG. 3, the invention thus provides an injection method which comprises an injection of a gaseous hydrogen/air mixture 12a that is richer in hydrogen than the stoichiometric level, into internal channel 6 of the injection device, and an injection of air 26a into external annular channel 8, so as to produce, at the outlet of said internal channel 6, a first flame front 30 resulting from rich combustion surrounded by a second flame front 31 resulting from lean combustion.

[0039] Internal channel 6 thus forms an injection tube for a rich mixture 12a of gaseous hydrogen/air and external annular channel 8 forms an injection tube for air 26a.

[0040] Rich mixture 12a of gaseous hydrogen and air is injected from an inlet 10 located at an upstream end of internal channel 6.

[0041] Internal channel 6 has an internal diameter d. The choice of internal diameter d of the channel depends on the desired thermal energy.

[0042] Returning to FIG. 2, a downstream end 16 of internal channel 6 is arranged upstream relative to a downstream end 24 of external annular channel 8. Downstream end 24 of external annular channel 8 is arranged at a distance r from downstream end 16 of internal channel 6, in the downstream direction. External annular channel 8 has an internal diameter D.

[0043] External annular channel 8 is configured to receive a second gas which is air 26a. This gas enters the external annular channel through an inlet 26a arranged at the upstream end of said external annular channel.

[0044] An annular swirler 28 is housed at said upstream end of external annular channel 8. This swirler may be radial or axial. Annular swirler 28 is arranged at a distance L from downstream end 36 of external annular channel 8. Air 26a passing through the external annular channel is set in rotation by external swirler 28. This generates a swirling system which will help to detach the second flame front from the outlet of the central channel.

[0045] Gaseous hydrogen/air premix 12a is injected into internal channel 6 formed by a tube creating a central injection duct. The premix richness level is greater than two, i.e. greater than 2 masses of hydrogen per 1 mass of air, and may even be greater than at least 4 in certain operating configurations.

[0046] Pure air 26a injected into annular channel 8 around internal channel 6 is injected in an amount calculated to target an overall injection richness level that is between 0.15 and 0.5, which amounts to a lean combustion overall. Pure air 26a is made to rotate within annular channel 8 by the axial or radial external swirler 28 located upstream of outlet plane 16a of downstream end 16 of the central duct for injecting the rich mixture of gaseous hydrogen/air.

[0047] Here, lip 16 of internal channel 6 is set back relative to outlet plane 24a of annular channel 8.

[0048] The operation of the injection device is described below, based on FIG. 3:

[0049] The injection of rich premix 12a of gaseous hydrogen/air into internal channel 6 makes it possible to create,

after ignition, a first flame front **30** at the outlet of internal channel **6**, resulting from the rich combustion of said mixture. This flame front attaches to lip **16** of internal channel **6**. This rich combustion, for example at a richness level that is greater than 2, is carried out with a flame front temperature that is lower than 1800 K so as not to generate nitrogen oxides. Flame front **30** is laminar and is not subject to diffusive-thermal instabilities due to a Lewis number that is greater than 1.

[0050] The injection of air **26a** at external annular channel **8** allows quickly diluting and confining the burned gases resulting from combustion of the rich premix. The presence of a strong turbulent shear layer makes it possible to reduce the local richness. This mixture is then burned and generates a second flame front **31** from lean combustion. This flame front remains stabilized due to the turbulent quenching caused by the provided air and due to the high reactivity of gaseous hydrogen despite the strong flame stretch. This second flame front resulting from lean combustion is also at a temperature below 1800 K, limiting the formation of nitrogen oxides. Flame front **31** is turbulent and is not attached to lip **16** of internal channel **6**. The length of the flame will depend on the conditions of entry of the fuels and oxidants and in particular on the momentum ratio, the setback of the internal channel relative to the external annular channel, and the presence of eddies in the flame.

[0051] The creation of these two flame fronts, rich flame **30** and lean flame **31**, makes it possible to distribute the thermal-acoustic load resulting from combustion over a larger surface area, and therefore to reduce the noise pollution resulting from combustion. Similarly, the stabilization of two flame fronts at the burner lips divides the thermal-acoustic loads related to combustion and reduces the generated noise.

[0052] The combustion method of this document thus achieves hydrogen combustion in stages in order to bypass the nitrogen oxide formation zone, by means of combustion of a rich premix of gaseous hydrogen/air in a first zone, internal flame **30**, and combustion of the residual gases in a second zone, flame **31** around flame **30**.

[0053] The risk of flashback is limited with rich combustion because the first flame front does not have diffusive-thermal instabilities. The speed of the first flame is thus not accelerated by instabilities.

[0054] Integrity of the combustion chamber is also ensured because, by carrying out combustions at high and low richness levels, the flame temperatures are lower than by carrying out combustion under stoichiometric conditions. Potential flame fronts originating from stoichiometric zones are not attached to the lips of the injector, which limits damage to the injector.

[0055] One exemplary embodiment provides, for operation under typical conditions of a gas turbine of a turboprop engine, a richness level in the rich zone that is on the order of 4 for the gaseous hydrogen/air mixture injected by internal channel **6**, i.e. a richness well above a stoichiometric richness level **1**, and air supplied by means of annular channel **8** in such an amount that the overall richness level is fixed between 0.15 and 0.50 depending on the operating points of the turboprop engine.

[0056] FIG. 4 shows various possible embodiments of the outlet of internal channel **6** for injecting the premix. The shape and thickness of outlet **16** of internal channel, **16a**, **16b**, **16c**, may be adjusted relative to the basic shape **16** of

the internal channel shown in FIG. 4(A). In FIG. 4(B), end **16a** of the internal channel is formed as an inward bevel, while in FIG. 4(C), end **16b** continues to flare outward in a bevel. In FIG. 4(D), end **16c** of the internal channel flares out but has an end face perpendicular to the longitudinal axis of the channel. These different configurations allow managing the attachment of first flame front **30** to lip **16** according to the configurations of the injection system.

[0057] In FIG. 4(E), a swirler **17** is added in internal channel **6** in order to homogenize the gaseous hydrogen/air premix.

[0058] As for external channel **8**, it may open out from a wall **240** as shown in FIG. 5, and the channel may have different outlet lip configurations:

[0059] Straight outlet **24** in FIG. 5(A), flaring conical inclined outlet **24a** in FIG. 5(B), or narrowing conical outlet **24b** as in FIG. 5(C). These different configurations allow adjusting the exit speed of the air surrounding the rich flame exiting internal channel **6**.

[0060] This disclosure thus relates to a method for injecting gaseous hydrogen premixed with air for an aeronautical gas turbine based on staged combustion, wherein:

- [0061]** a. combustion of the gaseous hydrogen/air premix at high richness takes place in a first region and generates a first flame front attached to the lips of the injector;
 - [0062]** b. rapid mixing of the combustion products via the injection of air in order to be burned in a second region while generating a stable second flame front.
- [0063]** This method allows in particular:
- [0064]** a. obtaining aerodynamically stabilized flames over a wide operating range,
 - [0065]** b. achieving combustion with very low nitrogen oxide emissions,
 - [0066]** c. avoiding the risk of flashback of the second flame front,
 - [0067]** d. reducing noise pollution related to hydrogen combustion,
 - [0068]** e. ensuring the integrity and service life of the injector.

[0069] The method as defined in the claims is not limited to the above description and in

[0070] particular may be applied to injection systems arranged in the rear walls of combustion chambers or extending beyond such walls.

1. An injection method, for an injection device in a combustion chamber of an aircraft turbine engine, said injection device comprising an internal channel surrounded by an external annular channel, said channels leading into said combustion chamber of said aircraft turbine engine, the method comprising:

an injection of a gaseous hydrogen/air mixture having a greater hydrogen richness level than the stoichiometric amount into said internal channel and

an injection of air into said external annular channel, producing at the an outlet of said internal channel, after ignition of said mixture at the outlet of the internal channel, a first flame front resulting from rich combustion, external to said internal channel and attached to a lip at the outlet of the internal channel,

said first flame front being surrounded by a second flame front resulting from lean combustion with the air exiting said external channel.

2. The injection method according to claim 1, wherein said gaseous hydrogen air mixture has a hydrogen richness level that is greater than 2.

3. The injection method according to claim 1, wherein said gaseous hydrogen/air mixture has a hydrogen richness level that is greater than or equal to 4.

4. The injection method according to claim 1, wherein an air flow rate in the external annular channel is chosen such that an overall richness level at the outlet of the injection device is set between 0.15 and 0.5 depending on the operating points of the turbine engine.

5. The injection method according to claim 1, wherein the injection of the gaseous hydrogen/air mixture and the device are configured to create, at the outlet of the internal channel, said first flame front resulting from a rich combustion of said mixture, and to attach it to a lip of the internal channel after ignition of said mixture.

6. The injection method according to claim 5, wherein the gaseous-hydrogen richness of the mixture is chosen so that

said rich combustion is carried out with a flame front temperature of less than 1800 K.

7. The injection method according to claim 5, wherein the gaseous-hydrogen richness of the mixture is chosen so that the first flame front is laminar and has a Lewis number greater than 1, limiting diffusive-thermal instabilities.

8. The injection method according to claim 5, wherein the mixture burned in the first flame front generates residual gases burned in the second flame front which is stabilized by the supply of air from the external annular channel.

9. The injection method according to claim 8, wherein the second flame front is maintained at a temperature below 1800 K.

10. The injection method according to claim 8, wherein the air injected by the annular channel is rotated by an annular swirler.

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